

Introduction to Databases

- COMP5211 - Week 1
- What a database management system does
- Why files on disk are not enough
- Roadmap: model, tables, keys, SQL, joins

The Relational Model

- Data is stored as relations, not pointers
- A relation is a set of tuples
- Every attribute has a domain
- Order of rows carries no meaning
- Proposed by E. F. Codd in 1970

Tables, Rows and Columns

- A table is one relation on disk
- A row is one tuple, one real-world fact
- A column is one attribute with one type
- NULL means unknown, not zero

Keys and Constraints

- A primary key identifies a row uniquely
- A foreign key points at another table's key
- Referential integrity forbids dangling references
- UNIQUE and NOT NULL are checked on write

SQL SELECT

- SELECT columns FROM table WHERE condition
- WHERE filters rows before grouping
- ORDER BY sorts the result, not the table
- LIMIT caps how many rows come back

Joins and Normalisation

- An INNER JOIN keeps only matching rows
- A LEFT JOIN keeps unmatched left rows too
- First normal form: no repeating groups
- Third normal form removes transitive dependencies
- Normalise to stop update anomalies